

Physics 827: Homework Set No. 1

Due date: Thursday, September 29, 2011, 3:00pm in the grader's office (see course web site)

Total point value of set: 55 points

Problem 1 (5 pts.): Exercise 1.1.2 (Shankar p.3)

Problem 2 (5 pts.): Exercise 1.1.3 (Shankar p.4)

Problem 3 (10 pts.): Orthonormalize the vectors $|I\rangle \leftrightarrow (0, 3, 4)$, $|II\rangle \leftrightarrow (2, 0, 0)$, $|III\rangle \leftrightarrow (0, -1, 3)$, using the Gram-Schmidt procedure and starting with $|I\rangle$.

Problem 4 (5 pts.): Exercise 1.3.4 (Shankar p.17, proof of triangle inequality)

Problem 5 (15 pts.): Exercise 1.4.1 (Shankar p.18)

Problem 6 (10 pts.): Exercise 1.4.2 (Shankar p.18)

Problem 7 (5 pts.): Exercise 1.6.1 (Shankar p.22)

Exercise 1.1.2 (p. 3)

HW 1

Physics 827
(QM1)

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The rule given for addition and scalar multiplication is that for the cartesian components of 3-vectors in $\mathbb{V}^3(\mathbb{R})$.

- The null vector is $|0\rangle = (0, 0, 0)$; it obviously satisfies $|V\rangle + |0\rangle = (v_1, v_2, v_3) + (0, 0, 0) = (v_1+0, v_2+0, v_3+0) = (v_1, v_2, v_3) = |V\rangle$
- The inverse of $|V\rangle = (a, b, c)$ is $|-V\rangle = (-a, -b, -c) = -(a, b, c) = -|V\rangle$
- The vectors of the form $(a, b, 1)$ do not include the null vector (ie. $\alpha(a, b, 1) \notin \mathbb{V}$ for $\alpha=0$), so $\mathbb{V} = \{(a, b, 1) | a, b \in \mathbb{R}\}$ is not a vector space.

Exercise 1.1.3 (p. 4)

- $\mathbb{V}_1 = \{f(x) | 0 \leq x \leq L, f(0) = f(L) = 0\}$

If I add two such functions or multiply such a function with a constant, the result will satisfy the boundary condition ✓

The null vector $f(x) \equiv 0 \forall x$ satisfies the boundary condition ✓

The inverse $-f(x)$ satisfies the boundary condition ✓

Distributive, commutative and associative properties are satisfied independent of boundary conditions ✓

$\Rightarrow \mathbb{V}_1$ is a vector space.

- $\mathbb{V}_2 = \{f(x) | 0 \leq x \leq L, f(0) = f(L)\}$ (periodic functions)

Adding two periodic functions or multiplying them by a constant results again in periodic functions ✓

The null function is periodic ✓

- The inverse - $f(x)$ of a periodic function is periodic ✓
- Distribution, commutative and associative properties hold independent of boundary conditions

$\Rightarrow V_2$ is a vector space

- $V_3 = \{ f(x) \mid 0 \leq x \leq L, f(0) = 4 \}$

This is not a vector space since $g(x) = \alpha f(x), \alpha \neq 1$, does not satisfy $g(0) = 4$.

• Exercise 1.3.2 (modified) (p. 16)

Orth. normalize the 3 vectors $|I\rangle = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$, $|II\rangle = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$, $|III\rangle = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$

$$|1\rangle = \frac{|I\rangle}{\sqrt{\langle I|I\rangle}} = \frac{1}{\sqrt{4+0+0}} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|2'\rangle = |II\rangle - |1\rangle \underbrace{\langle 1|II\rangle}_0 = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} \quad \left(\text{since } |I\rangle \text{ and } |II\rangle \text{ are already orthogonal} \right)$$

$$|2\rangle = \frac{|2'\rangle}{\sqrt{\langle 2'|2'\rangle}} = \frac{1}{\sqrt{0+9+16}} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix}$$

$$\begin{aligned} |3'\rangle &= |III\rangle - |1\rangle \underbrace{\langle 1|III\rangle}_0 - |2\rangle \langle 2|III\rangle = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix} \left(0 - \frac{3}{5} + \frac{12}{5} \right) \\ &= \begin{bmatrix} 0 \\ -1 - \frac{27}{25} \\ 3 - \frac{36}{5} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{52}{25} \\ +\frac{39}{25} \end{bmatrix} \end{aligned}$$

$$|3\rangle = \frac{|3'\rangle}{\sqrt{\langle 3'|3'\rangle}} = \begin{bmatrix} 0 \\ -\frac{52}{25} \\ +\frac{39}{25} \end{bmatrix} \frac{1}{\frac{1}{25} \sqrt{0 + 4^2 \cdot 13^2 + 3^2 \cdot 13^2}} = \begin{bmatrix} 0 \\ -52 \\ +39 \end{bmatrix} \frac{1}{13 \cdot 5} = \begin{bmatrix} 0 \\ -4/5 \\ 3/5 \end{bmatrix}$$

• Exercise 1.3.4 (p. 17)

Prove $|V+W| \leq |V| + |W|$.

Proof: Since both sides are real and positive,
it's sufficient to prove $|V+W|^2 \leq (|V|+|W|)^2$
or $|V|^2 + 2|V||W| + |W|^2 - \langle V+W|V+W \rangle \geq 0$

Work on l.h.s.:

$$\begin{aligned} & |V|^2 + 2|V||W| + |W|^2 - \underbrace{\langle V|V \rangle}_{|V|^2} - \langle V|W \rangle - \underbrace{\langle W|V \rangle}_{=\langle V|W \rangle^*} - \underbrace{\langle W|W \rangle}_{|W|^2} \\ &= 2(|V||W| - \underbrace{\operatorname{Re} \langle V|W \rangle}_{\leq |\langle V|W \rangle|}) \end{aligned}$$

$$\begin{aligned} \text{Since } |\langle V|W \rangle|^2 &= (\operatorname{Re} \langle V|W \rangle)^2 + (\operatorname{Im} \langle V|W \rangle)^2 \\ &\geq (\operatorname{Re} \langle V|W \rangle)^2 \end{aligned}$$

$$\geq 2(|V||W| - |\langle V|W \rangle|) \geq 0 \quad \uparrow \text{Schwarz inequality} \quad \diamond$$

• Exercise 1.4.1 (p. 18)

In $\mathbb{V}^n$, prove that $\mathbb{V}_\perp^{n-1} = \{ |W\rangle \mid \langle V|W \rangle = 0, |V\rangle \neq 0 \}$
is a subspace of dimension $n-1$.

Proof: Since $\mathbb{V}^n$ has dimension n , we can find n linearly independent vectors in $\mathbb{V}^n$, $\{|W_i\rangle \mid i=1, \dots, n\}$. Using the Gram-Schmidt algorithm, we can orthonormalize these vectors to obtain an orthonormal basis

$\{|i\rangle | i=1, \dots, n\}$. Now choose an arbitrary vector $|V\rangle \neq |0\rangle$ in V^n . We can decompose it in this basis as

$$|V\rangle = \sum_{i=1}^n v_i |i\rangle, \quad (*)$$

where at least one of the components v_i is non-zero.

Let us assume without loss of generality that $v_1 \neq 0$.

(We can always achieve that by renumbering the basis states accordingly.) Eq. (*) can then be rewritten as

$$|V\rangle = v_1 |1\rangle + \sum_{i=2}^n v_i |i\rangle \Rightarrow |1\rangle = \sum_{i=2}^n \frac{v_i}{v_1} |i\rangle - \frac{1}{v_1} |V\rangle \quad (**)$$

Since the basis was orthonormal, $|1\rangle$ is orthogonal to all states $|i\rangle, i=2, \dots, n$. For (**) to hold, $|V\rangle$ must therefore have a part that is orthogonal to all states $|i\rangle, i=2, \dots, n$. Hence $|V\rangle$ is linearly independent of $\{|i\rangle | i=2, \dots, n\}$, since the part of $|V\rangle$ that is orthogonal to these vectors cannot be obtained from any linear superposition of these vectors.

Let us now look at this new set of n linearly independent vectors given by $\{|V\rangle, |2\rangle, \dots, |n\rangle\}$.

Using Gram-Schmidt again, starting from $|V\rangle$, we obtain a new orthonormal basis of V^n ,

$\{|1'\rangle, |2'\rangle, \dots, |n'\rangle\}$ whose first unit vector $|1'\rangle = \frac{|V\rangle}{\sqrt{\langle V|V\rangle}}$ is in the direction of $|V\rangle$.

We can now decompose any vector $|W\rangle \in \mathbb{V}^n$ as

$$|W\rangle = \sum_{i=1}^n \omega_i |i'\rangle.$$

Any element of the set $\mathbb{V}_\perp \equiv \{|V_\perp\rangle \in \mathbb{V}^n, \langle V_\perp | V \rangle = 0\}$ of vectors $|V_\perp\rangle$ that are orthogonal to $|V\rangle$ are orthogonal to $|1'\rangle$ and thus can be written as

$$|V_\perp\rangle = \sum_{i=2}^n v_i |i'\rangle$$

with no component along $|1'\rangle$. The set $\{|i'\rangle | i=2, \dots, n\}$ forms an $(n-1)$ -dimensional orthonormal basis, so

the set $\mathbb{V}_\perp \equiv \mathbb{V}_\perp^{n-1}$ has dimension $n-1$. (We need to show that this subset of $\mathbb{V}^n$ is a vector space.) It contains

the null vector $|0\rangle$ since $\langle 0 | V \rangle = 0$ for any $|V\rangle \in \mathbb{V}^n$.

For any vector $|V_\perp\rangle \in \mathbb{V}_\perp^{n-1}$, its inverse $|-V_\perp\rangle$ is also in $\mathbb{V}_\perp^{n-1}$ since $\langle -V_\perp | V \rangle = -\langle V_\perp | V \rangle = 0$, so

$|-V_\perp\rangle$ is orthogonal to $|V\rangle$. Adding two vectors that are orthogonal to $|V\rangle$ or multiplying them by a constant again produces vectors that are orthogonal to $|V\rangle$, i.e. elements of $\mathbb{V}_\perp^{n-1}$.

Hence $\mathbb{V}_\perp^{n-1}$ satisfies all axioms that define a vector space; it is an $(n-1)$ -dimensional subspace of $\mathbb{V}^n$.

Exercise 1.4.2 (p. 18)

- The elements of $V_1^{n_1}$ can be written as

$$|V\rangle = \sum_{i=1}^{n_1} v_i |i\rangle$$

where $\{|i\rangle, i=1, \dots, n_1\}$ is an orthonormal basis of $V_1^{n_1}$.

- The elements of $V_2^{n_2}$ can be written as

$$|W\rangle = \sum_{i=1}^{n_2} w_i |i'\rangle$$

where $\{|i'\rangle, i=1, \dots, n_2\}$ is an orthonormal basis of $V_2^{n_2}$.

Since all elements of $V_1^{n_1}$ are orthogonal to all elements of $V_2^{n_2}$, all states $|i\rangle$ ($i=1, \dots, n_1$) are orthogonal to all states $|i'\rangle$ ($i=1, \dots, n_2$).

Hence the set $\{|i\rangle, |j'\rangle, i=1, \dots, n_1; j=1, \dots, n_2\}$ is an orthonormal basis for an (n_1+n_2) -dimensional vector space.

By definition, the elements of $V_1^{n_1} \oplus V_2^{n_2}$ are all linear combinations

$$|Z\rangle = a|V\rangle + b|W\rangle$$

$$|V\rangle \in V_1^{n_1}, |W\rangle \in V_2^{n_2}, \\ a, b \text{ constants.}$$

These can be written as

$$|Z\rangle = a \sum_{i=1}^{n_1} v_i |i\rangle + b \sum_{j=1}^{n_2} w_j |j'\rangle = \sum_{i=1}^{n_1} (av_i) |i\rangle + \sum_{j=1}^{n_2} (bw_j) |j'\rangle$$

i.e. $|Z\rangle$ can be decomposed in the orthonormal

basis set $\{|i\rangle, |j\rangle\}$ $i=1, \dots, n_1; j=1, \dots, n_2$ which is n_1+n_2 dimensional. Hence the dimensionality of $V_1^{n_1} \oplus V_2^{n_2}$ is not larger than n_1+n_2 .

Since the ^{orthogonal} basis elements themselves are also elements of $V_1^{n_1} \oplus V_2^{n_2}$, its dimensionality is at least n_1+n_2 . Hence $V_1^{n_1} \oplus V_2^{n_2}$ has dimensionality equal to n_1+n_2 .

Exercise 1.6.1 (p. 22)

From the columns of the matrix $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = [\Omega_{ij}]$ we read off the action of $\hat{\Omega}$ on the basis states:

$$\hat{\Omega}|1\rangle = |2\rangle, \quad \hat{\Omega}|2\rangle = |3\rangle, \quad \hat{\Omega}|3\rangle = |1\rangle.$$

So $\hat{\Omega}$ cyclically permutes the states $|1\rangle, |2\rangle, |3\rangle$.

This corresponds to a rotation by 120° ($\frac{1}{3}$ of 360°) around the axis $\frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}}$. ($\frac{1}{\sqrt{3}}$ normalizes

the unit vector pointing along this axis.)